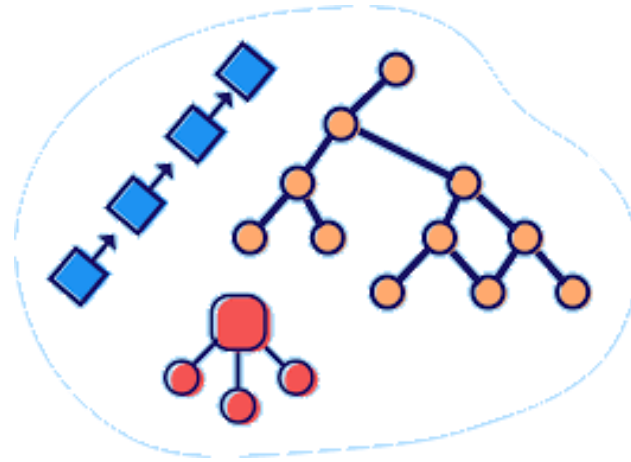


Introduction to Data Structures & Algorithms

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Introduction

- What is a data structure?
- What is an algorithm?
- Use of data structures & algorithms.
- Other benefits.

Data Structures

- A data structure is a way of storing and organizing data in a computer so that it can be accessed and manipulated efficiently.

Types of Data Structures

- Simple data structures
- Abstract data structures

Simple Data Structures

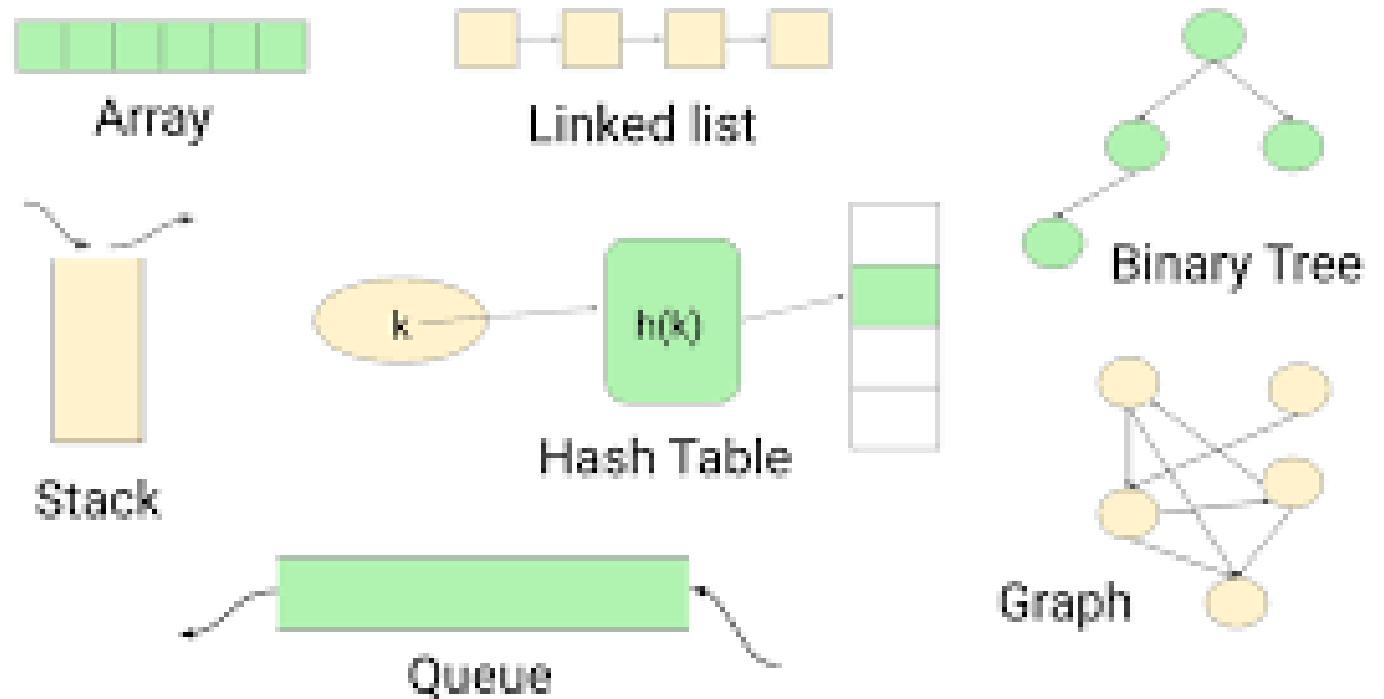
- Simply keep one or set of values without any special arrangement.
- Examples
 - Simple variable
 - Array
- Built in available with a programming languages.

Abstract Data Structures

- Abstract Data Structures (ADS) are models that define the properties of data and operations on that data without specifying the implementation details.
- They provide a high-level description of how data can be organized, accessed, and manipulated.
- Programmers have defined some ADSs having common usages.

Abstract Data Structures - Classification

- Linear
 - Linked list
 - Stack
 - Queue
- Non-linear
 - Tree
 - Graph
- Other
 - Hash

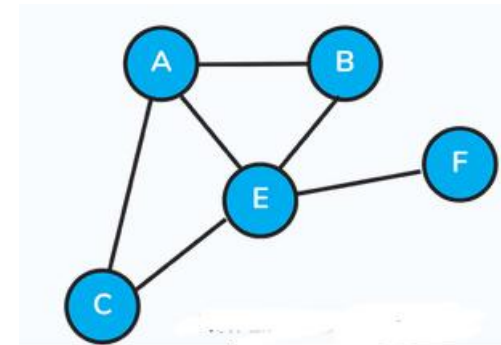
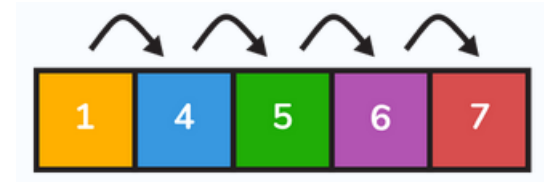


Abstract Data Structures – How?

- Some abstract data structures are built on available with programming languages.
 - Examples
 - Java list, set, map, etc..
- We can create our own data structures with specific required features.

Abstract Data Structures - Classification

- Linear data structures
 - Linked list
 - Stack
 - Queue
- Non-linear data structures
 - Tree
 - Binary search tree
 - Binary heap
 - Graph
- Other
 - Hash tables



Static Vs Dynamic Data Structures

Feature	Static Data Structure	Dynamic Data Structure
Size	Fixed size	Size can grow or shrink as needed
Flexibility	Less flexible	More flexible
Memory Allocation	Fixed at compile-time	Allocated at run-time
Memory Usage	May lead to unused memory (wastage)	Efficient memory use (allocated as needed)
Speed	Generally faster (no allocation/deallocation overhead)	Slightly slower (due to dynamic allocation/deallocation)
Resizing	Not possible without creating a new structure	Resizing possible at runtime
Examples	Arrays, Static stacks/queues	Linked lists, Dynamic stacks/queues, Trees, Graphs

Features of a Good Data Structure

- Abstract
- Reusable
- Efficient
- Flexible

Algorithms

Algorithms

- An algorithm is a step-by-step procedure for solving a problem or accomplishing a task.
- An algorithm may use specific data structures to help solving the problem.



Useful Algorithms - Examples

- Searching algorithms
- Sorting algorithms
- Traversing algorithms
- Optimization algorithms
- Etc..

Features of a Good Algorithm

- Effective
 - Does the work
- Efficient
 - Time
 - Resource
- Reusable
 - General

When you develop software , represent the data in the best possible data structures and process using the most effective and efficient algorithms.



Past Papers

2021 #1-b-i

Discuss two (02) main differences between static and dynamic data structures.

Answer

Static data structure

the size is allocated at the compile time. Therefore, the maximum size is fixed.

Dynamic data structure

the size is allocated at the run time. Therefore, the maximum size is flexible

2021 #1-b-ii

When selecting a suitable data structure for a particular application, the following characteristics are advantageous:

- Efficiency
- Reusability
- Abstraction

Discuss how the above characteristics are advantageous in a client-side implementation.

Answer

Efficiency: It makes the program very efficient in terms of time and space.

Reusability: Provides reusability means that multiple client programs can use the data structure.

Abstraction: this provides the level of abstraction. The client cannot see the internal working of the data structure, so it does not have to worry about the implementation part. The client can only see the interface.

Thank You

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